Lake Erie HAB-OFS Bulletin Guide

Created: May 1, 2017

Harmful algal blooms (HABs) of the cyanobacteria *Microcystis sp.* (commonly called "blue-green algae") occur nearly every year in Lake Erie and cause potential public health, economic and ecological impacts. The National Oceanic and Atmospheric Administration (NOAA) provides the HAB Operational Forecast System (HAB-OFS) Bulletin as a decision support tool. HAB-OFS Bulletins are emailed to subscribers during the Lake Erie HAB season and are viewable as georeferenced pdfs (see GeoPDF Viewing Instructions). To subscribe, click here. Please send your questions to hab@noaa.gov.

A) Bloom Analysis

This section provides the current and predicted *Microcystis* conditions as follows:

- **Bloom location, density, and toxicity:** The bloom location is described based on satellite imagery. NOAA partner observations are used to determine toxicity of the bloom.
- **Transport forecast:** Potential bloom transport is forecast using satellite imagery that is incorporated with modeled surface currents using a 2D particle trajectory model.
- **Mixing nowcast/forecast:** Vertical mixing of cyanobacterial cells through the water column reduces the visibility of the bloom at the surface. Forecasts of mixing and surface scum formation for three days following the bulletin date, are based on predicted winds. Nowcasts from imagery date to bulletin date are based on observed winds. Forecasts for mixing and surface scum formation for the seventy-two hours following the bulletin are based on predicted winds.
- **Potential decline:** The potential for bloom decline is forecast when temperatures decrease below 15°C.

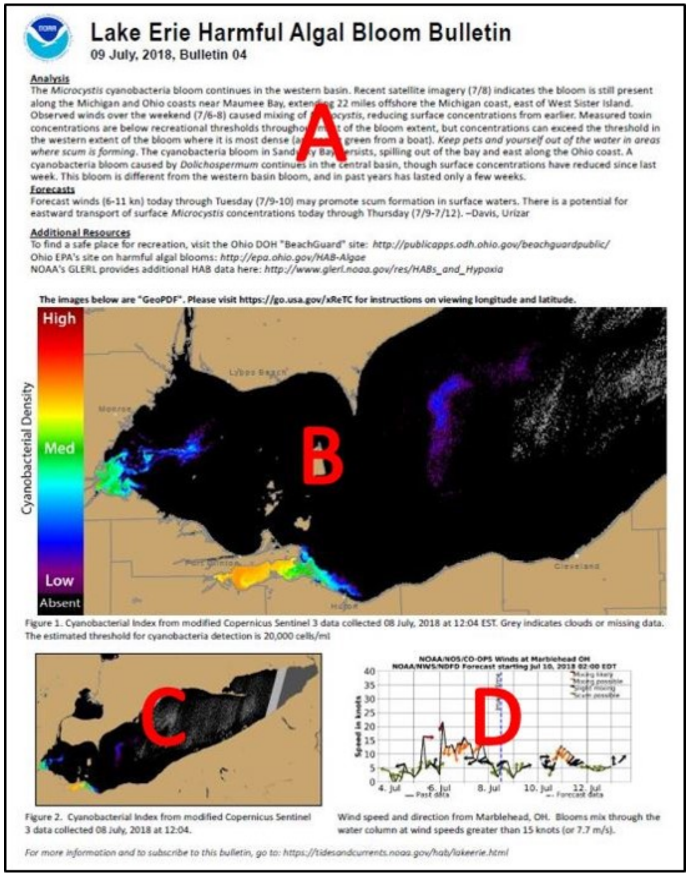
B) and C) Lake Erie Satellite Imagery

Georeferenced satellite images come from the Moderate Resolution Imaging Spectroradiometer (MODIS) sensors aboard the Terra and Aqua satellites and the Ocean and Land Color Instrument (OLCI) sensor aboard the Sentinel-3 satellite (click here for the Sentinel data usage legal notice).

Bloom density is processed using a cyanobacterial index represented on a color spectrum where purple represents the lowest density of cyanobacteria and red represents the highest. Black pixels indicate a bloom was not detected and gray indicates the presence of clouds.

Image B Western/central basins of Lake Erie where blooms tend to originate.

Image C Full extent of Lake Erie.



Lake Erie Harmful Algal Bloom Bulletin

09 July, 2018, Bulletin 04

Analysis

The *Microcystis* cyanobacteria bloom continues in the western basin. Recent satellite imagery (7/8) indicates the bloom is still present along the Michigan and Ohio coasts near Maumee Bay, extending 22 miles offshore the Michigan coast, east of West Sister Island. Observed winds over the weekend (7/6-8) caused mixing of [illegible]ocystis, reducing surface concentrations from earlier. Measured toxin concentrations are below recreational thresholds througho[illegible]t of the bloom extent, but concentrations can exceed the threshold in the western extent of the bloom where it is most dense (a[illegible] green from a boat). *Keep pets and yourself out of the water in areas where scum is forming*. The cyanobacteria bloom in Sand[illegible]y Bay [illegible]ersists, spilling out of the bay and east along the Ohio coast. A cyanobacteria bloom caused by *Dolichospermum* continues in the central basin, though surface concentrations have reduced since last week. This bloom is different from the western basin bloom, and in past years has lasted only a few weeks.

Forecasts

Forecast winds (6-11 kn) today through Tuesday (7/9-10) may promote scum formation in surface waters. There is a potential for eastward transport of surface *Microcystis* concentrations today through Thursday (7/9-7/12). –Davis, Urizar

Additional Resources

To find a safe place for recreation, visit the Ohio DOH "BeachGuard" site: *http://publicapps.odh.ohio.gov/beachguardpublic/*
Ohio EPA's site on harmful algal blooms: *http://epa.ohio.gov/HAB-Algae*
NOAA's GLERL provides additional HAB data here: *http://www.glerl.noaa.gov/res/HABs_and_Hypoxia*

The images below are "GeoPDF". Please visit https://go.usa.gov/xReTC for instructions on viewing longitude and latitude.

Figure 1. Cyanobacterial Index from modified Copernicus Sentinel 3 data collected 08 July, 2018 at 12:04 EST. Grey indicates clouds or missing data. The estimated threshold for cyanobacteria detection is 20,000 cells/ml

Figure 2. Cyanobacterial Index from modified Copernicus Sentinel 3 data collected 08 July, 2018 at 12:04.

Wind speed and direction from Marblehead, OH. Blooms mix through the water column at wind speeds greater than 15 knots (or 7.7 m/s).

For more information and to subscribe to this bulletin, go to: https://tidesandcurrents.noaa.gov/hab/lakeerie.html

D) Mixing Forecast

A graphical representation of the potential for mixing based on wind speed.

- ⇨ <5 knots winds —Potential for scum formation
- ⇨ 5-10 knots winds —Slight potential for mixing
- ⇨ 10-15 knots—Increased potential for mixing
- ⇨ 15+ knots winds —Mixing is likely

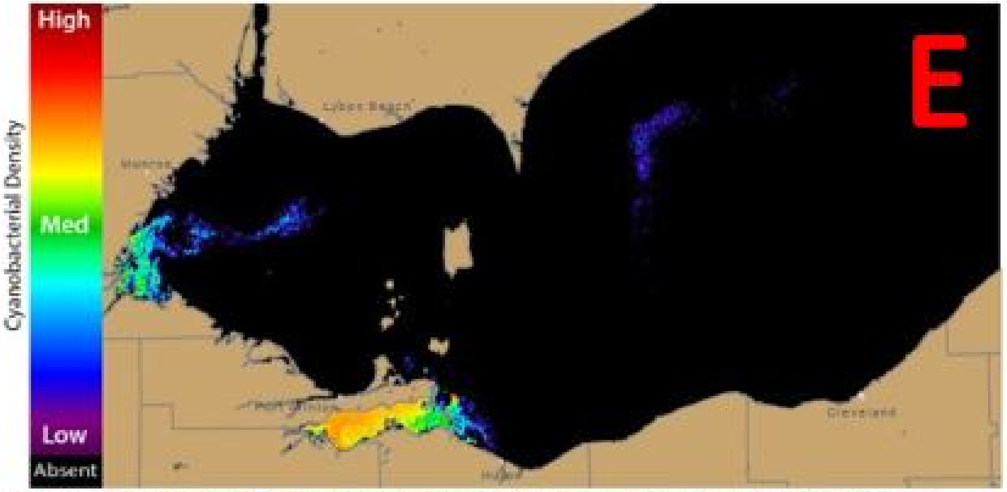

High
Med
Low
Absent
Cyanobacterial Density
E
Monroe
Port Clinton
Cleveland
Huron

Figure 3. Nowcast position of bloom for 09 July, 2018 using LEOFS modelled currents to move the bloom from the 08 July, 2018 image.

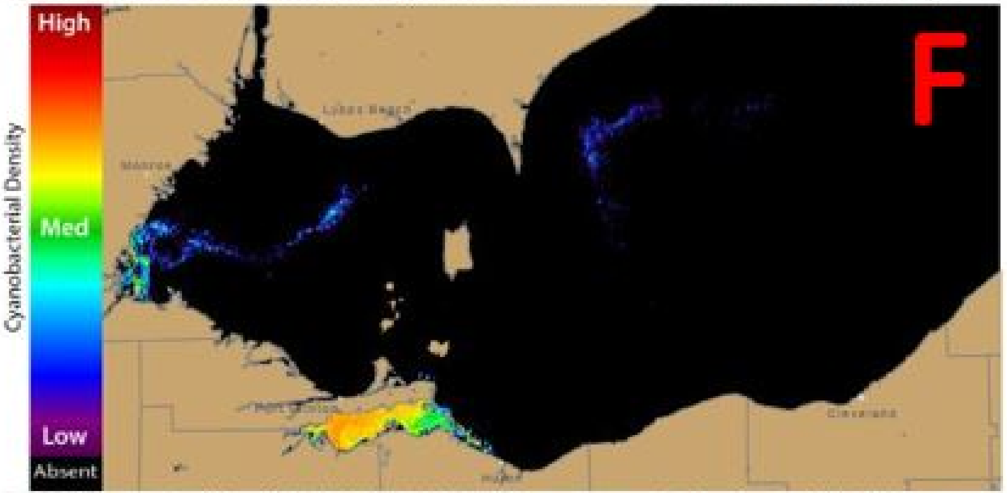
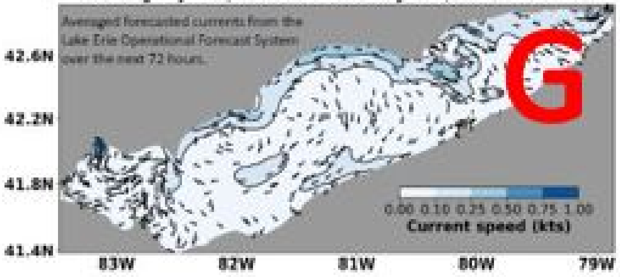

High
Med
Low
Absent
Cyanobacterial Density
F
Monroe
Port Clinton
Cleveland
Huron

Figure 4. Forecast position of bloom for 12 July, 2018 using LEOFS modelled currents to move the bloom from the 08 July, 2018 image.

FVCOM Currents 72-hr Average
starting at Jul 09, 2018 00:00 UTC to Jul 12, 2018 00:00 UTC


Averaged forecasted currents from the
Lake Erie Operational Forecast System
over the next 72 hours.
G
42.6N
42.2N
41.8N
41.4N
83W
82W
81W
80W
79W
0.00 0.10 0.25 0.50 0.75 1.00
Current speed (kts)

For more information and to subscribe, please visit the NOAA HAB Forecast page: https://tidesandcurrents.noaa.gov/hab/lakeerie.html

E) Transport Nowcast and F) Transport Forecast

Transport is predicted by incorporating satellite imagery and surface currents from the Lake Erie Operational Forecast System (LEOFS) into the General NOAA Operational Modeling Environment (GNOME) particle trajectory model.

- A transport nowcast predicts bloom movement from the date of satellite imagery to date of the bulletin, and is based on wind and current forecasts from the LEOFS.
- A transport forecast predicts bloom movement over 72 hours from the bulletin date, and is based on wind and current forecasts from the LEOFS.

G) Currents

Image depicts the average forecasted currents data over 72 hours from the date of the bulletin, and is based on winds data from the LEOFS.

NOAA/National Ocean Service

For more information and to subscribe, please visit the NOAA HAB Forecast page by clicking here.

Email: hab@noaa.gov